

Question Bank — Digital Logic + COA

Module 5 companion | PYQ-style + Practice | exam-tagged, answers & explanations

■ How to use this bank

Two sections: PYQ-style (modelled on real IBPS/SSC/GATE/PSU/DSSSB questions) and **Practice** (full coverage by sub-topic).

Every question has answer + explanation. **Tags:** IBPS • Coal India • DSSSB • All 3 • GATE-level • SSC/other.

For IBPS: number systems, gates, RAM/ROM, memory hierarchy are enough; GATE-tagged (K-maps, flip-flop design, cache math, pipelining) are Coal-India depth.

SECTION A — PYQ-style (modelled on actual exam questions)

Number systems

Q1

All 3 exams

The binary equivalent of decimal 13 is:

- (a) 1101
- (b) 1011
- (c) 1110
- (d) 1001

Answer: (a)

$13 = 8+4+1 = 1101$ in binary.

Q2

All 3 exams

SSC/other

The hexadecimal equivalent of binary 11111111 is:

- (a) FF
- (b) EE
- (c) F0
- (d) 0F

Answer: (a)

$11111111 = \text{two nibbles } 1111 \ 1111 = F \ F = FF \ (255).$

Q3

IBPS SO IT

All 3 exams

One byte equals how many bits?

- (a) 4
- (b) 8
- (c) 16
- (d) 2

Answer: (b)

1 byte = 8 bits; 1 nibble = 4 bits.

Q4

All 3 exams

The octal number system uses digits:

- (a) 0-7
- (b) 0-8
- (c) 0-9
- (d) 0-F

Answer: (a)

Octal is base 8, digits 0-7.

Q5

Coal India MT

GATE-level

The 2's complement of binary 0101 (4-bit) is:

- (a) 1010
- (b) 1011
- (c) 1101
- (d) 0110

Answer: (b)

Invert 0101 → 1010, add 1 → 1011 (represents -5).

Q6

GATE-level

Coal India MT

The range of an 8-bit signed (2's complement) number is:

- (a) 0 to 255
- (b) -128 to 127
- (c) -127 to 128
- (d) -255 to 255

Answer: (b)

n-bit 2's complement range = $-2^{(n-1)}$ to $2^{(n-1)}-1 = -128$ to 127.

Boolean algebra & gates

Q7

All 3 exams

Which gates are known as universal gates?

- (a) AND, OR
- (b) NAND, NOR
- (c) XOR, XNOR
- (d) NOT, AND

Answer: (b)

NAND and NOR are universal — any circuit can be built from only NANDs or only NORs.

Q8

All 3 exams

SSC/other

The output of an XOR gate is 1 when:

- (a) Both inputs equal
- (b) Inputs differ
- (c) Both are 1
- (d) Both are 0

Answer: (b)

XOR outputs 1 when inputs differ; XNOR outputs 1 when they are equal.

Q9

All 3 exams

De Morgan's law states that $(A \cdot B)'$ equals:

- (a) $A' \cdot B'$
- (b) $A' + B'$
- (c) $A + B$
- (d) AB

Answer: (b)

$(A \cdot B)' = A' + B'$. Also $(A + B)' = A' \cdot B'$.

Q10

All 3 exams

A NOT gate is also called a(n):

- (a) Buffer
- (b) Inverter
- (c) Adder
- (d) Latch

Answer: (b)

A NOT gate inverts its input; hence 'inverter'.

Q11

DSSSB TGT

All 3 exams

How many rows does the truth table of a 3-input gate have?

- (a) 3
- (b) 6
- (c) 8
- (d) 9

Answer: (c)

2^n rows; $2^3 = 8$.

Q12

GATE-level

Coal India MT

K-maps are used to:

- (a) Store data
- (b) Simplify Boolean expressions
- (c) Schedule the CPU
- (d) Route packets

Answer: (b)

Karnaugh maps minimise Boolean expressions (up to ~4 variables). GATE-depth.

Circuits

Q13

All 3 exams

A circuit whose output depends only on present inputs is:

- (a) Sequential
- (b) Combinational
- (c) Memory
- (d) Clocked

Answer: (b)

Combinational logic depends only on current inputs (adder, MUX, decoder).

Q14

All 3 exams

SSC/other

A flip-flop stores:

- (a) 8 bits
- (b) 1 bit
- (c) 1 byte
- (d) 0 bits

Answer: (b)

A flip-flop is a 1-bit memory element.

Q15

IBPS SO IT

All 3 exams

A multiplexer (MUX):

- (a) Splits one input to many outputs
- (b) Selects one of many inputs to a single output
- (c) Stores data
- (d) Adds numbers

Answer: (b)

A MUX routes one of several inputs to the output based on select lines; a DEMUX does the reverse.

Q16

Coal India MT

GATE-level

A circuit whose output depends on inputs AND past state is:

- (a) Combinational
- (b) Sequential
- (c) Purely resistive
- (d) Asynchronous only

Answer: (b)

Sequential circuits use memory (flip-flops) so output depends on inputs + stored state.

CPU & buses

Q17

All 3 exams

In the von Neumann architecture, instructions and data share:

- (a) Separate memories
- (b) The same memory
- (c) No memory
- (d) Only registers

Answer: (b)

Von Neumann uses one shared memory for instructions and data (the 'von Neumann bottleneck').

Q18

All 3 exams

SSC/other

The ALU performs:

- (a) Control signals
- (b) Arithmetic and logic operations
- (c) Memory storage
- (d) I/O only

Answer: (b)

The Arithmetic Logic Unit does arithmetic and logic; the control unit directs operations.

Q19

IBPS SO IT

All 3 exams

The register holding the address of the next instruction is the:

- (a) IR
- (b) PC (Program Counter)
- (c) MAR
- (d) Accumulator

Answer: (b)

The Program Counter holds the next instruction's address; the IR holds the current instruction.

Q20

All 3 exams

The address bus is:

- (a) Bidirectional
- (b) Unidirectional
- (c) Optional
- (d) Same as data bus

Answer: (b)

The address bus is unidirectional (CPU→memory); the data bus is bidirectional.

Q21

GATE-level

Coal India MT

If the address bus is 16 bits wide, the addressable memory is:

- (a) 16 bytes
- (b) 64 KB
- (c) 16 KB
- (d) 1 MB

Answer: (b)

$2^{16} = 65536 = 64 \text{ KB}$ addressable locations.

Q22

Coal India MT

GATE-level

Harvard architecture differs from von Neumann by having:

- (a) One memory
- (b) Separate instruction and data memories/buses
- (c) No ALU
- (d) No registers

Answer: (b)

Harvard uses separate memory and buses for instructions and data (faster; used in DSP/microcontrollers).

Memory & I/O

Q23

All 3 exams

Which memory is volatile?

- (a) ROM
- (b) RAM
- (c) Hard disk
- (d) EEPROM

Answer: (b)

RAM loses its contents on power-off (volatile); ROM is non-volatile.

Q24

All 3 exams

SSC/other

Which is fastest in the memory hierarchy?

- (a) Hard disk
- (b) Main memory
- (c) Cache
- (d) Registers

Answer: (d)

Registers (inside the CPU) are the fastest; then cache, RAM, disk.

Q25

IBPS SO IT

All 3 exams

Cache memory is typically made of:

- (a) DRAM
- (b) SRAM
- (c) Flash
- (d) Magnetic tape

Answer: (b)

Cache uses fast SRAM; main memory uses denser DRAM.

Q26

All 3 exams

BIOS/firmware is typically stored in:

- (a) RAM
- (b) ROM
- (c) Cache
- (d) Registers

Answer: (b)

Firmware/BIOS is stored in non-volatile ROM (often EEPROM/flash).

Q27

DSSSB TGT

All 3 exams

Which enables data transfer between device and memory without CPU involvement for the bulk transfer?

- (a) Programmed I/O
- (b) Interrupt I/O
- (c) DMA
- (d) Polling

Answer: (c)

DMA (Direct Memory Access) transfers data directly between device and memory, freeing the CPU.

Q28

GATE-level

Coal India MT

Cache works because programs exhibit:

- (a) Randomness
- (b) Locality of reference
- (c) Fragmentation
- (d) Recursion

Answer: (b)

Temporal and spatial locality make caching effective. GATE-depth topics include mapping and hit-ratio math.

SECTION B — Practice (full syllabus coverage for revision)

B1. Number systems

Q29

All 3 exams

Decimal 255 in binary is:

- (a) 11111111
- (b) 10000000
- (c) 11110000
- (d) 11111110

Answer: (a) $255 = 2^8 - 1 = 11111111.$ **Q30**

All 3 exams

Binary 1010 in decimal is:

- (a) 8
- (b) 10
- (c) 12
- (d) 5

Answer: (b) $1010 = 8 + 0 + 2 + 0 = 10.$ **Q31**

All 3 exams

To convert binary to hexadecimal, group bits in:

- (a) 2s
- (b) 3s
- (c) 4s
- (d) 8s

Answer: (c)

Group binary digits in 4s for hex (3s for octal).

Q32

GATE-level

Coal India MT

The 1's complement of 1100 is:

- (a) 0011
- (b) 0100
- (c) 1011
- (d) 1101

Answer: (a)1's complement inverts each bit: $1100 \rightarrow 0011.$ **Q33**

DSSSB TGT

All 3 exams

ASCII originally used how many bits per character?

- (a) 7
- (b) 8
- (c) 16
- (d) 32

Answer: (a)

Standard ASCII is 7-bit (extended ASCII uses 8); Unicode is multi-byte.

B2. Gates & circuits

Q34

All 3 exams

The output of an AND gate is 1 only when:

- (a) Any input is 1
- (b) All inputs are 1
- (c) All inputs are 0
- (d) Inputs differ

Answer: (b)

AND outputs 1 only when all inputs are 1.

Q35

All 3 exams

An OR gate outputs 0 only when:

- (a) Any input is 1
- (b) All inputs are 0
- (c) All inputs are 1
- (d) Inputs differ

Answer: (b)

OR outputs 0 only when every input is 0.

Q36

All 3 exams

A decoder does the opposite of a(n):

- (a) MUX
- (b) Encoder
- (c) Adder
- (d) Flip-flop

Answer: (b)

A decoder expands n inputs to 2^n outputs; an encoder does the reverse.

Q37

All 3 exams

A half adder adds:

- (a) Two bits (no carry-in)
- (b) Three bits
- (c) Two bytes
- (d) With carry-in

Answer: (a)

A half adder adds two bits (sum + carry); a full adder includes a carry-in.

Q38

GATE-level

Coal India MT

A JK flip-flop with $J=K=1$ will:

- (a) Hold
- (b) Reset
- (c) Set
- (d) Toggle

Answer: (d)

With $J=K=1$, a JK flip-flop toggles its output on each clock.

B3. CPU & instruction cycle

Q39

All 3 exams

The basic instruction cycle is:

- (a) Fetch-Decode-Execute
- (b) Read-Write-Store
- (c) Load-Save-Print
- (d) Input-Output-Loop

Answer: (a)

The CPU repeats Fetch → Decode → Execute (with operand fetch/interrupt checks).

Q40

All 3 exams

The control unit's job is to:

- (a) Do arithmetic
- (b) Generate control signals to direct operations
- (c) Store data
- (d) Handle I/O ports

Answer: (b)

The control unit decodes instructions and issues control signals.

Q41

All 3 exams

In immediate addressing mode, the operand is:

- (a) In memory
- (b) In a register
- (c) A constant in the instruction
- (d) At a pointer

Answer: (c)

Immediate mode carries the operand as a constant inside the instruction.

Q42

GATE-level

Coal India MT

In indirect addressing, the instruction gives:

- (a) The operand
- (b) The address of the operand's address
- (c) A register
- (d) A constant

Answer: (b)

Indirect addressing points to a location that itself holds the operand's address.

B4. Memory & I/O

Q43

All 3 exams

SRAM differs from DRAM in that SRAM:

- (a) Needs refreshing
- (b) Is faster and used for cache
- (c) Is denser
- (d) Is non-volatile

Answer: (b)

SRAM is faster (used for cache) and needs no refresh; DRAM is denser (main memory) but must refresh.

Q44

All 3 exams

EEPROM is:

- (a) Volatile RAM
- (b) Electrically erasable non-volatile ROM
- (c) Cache
- (d) A register

Answer: (b)

EEPROM is electrically erasable/reprogrammable non-volatile memory.

Q45

All 3 exams

A cache 'hit' means the requested data is:

- (a) Not in cache
- (b) Found in cache
- (c) On disk
- (d) In a register

Answer: (b)

A hit = data found in cache; a miss = fetch from a lower level.

Q46

All 3 exams

Programmed I/O wastes CPU time because the CPU must:

- (a) Sleep
- (b) Continuously poll the device
- (c) Use DMA
- (d) Interrupt itself

Answer: (b)

In programmed I/O the CPU polls (busy-waits) the device status, wasting cycles.

Q47

GATE-level

Coal India MT

Direct-mapped cache maps a memory block to:

- (a) Any line
- (b) Exactly one line
- (c) Two lines
- (d) A set of lines

Answer: (b)

Direct mapping sends each block to exactly one cache line; set-associative maps to a set; fully associative to any line.

Coverage & next

✓ What this bank covered

PYQ + Practice across Digital Logic (number systems, complements, gates, circuits) and COA (CPU, buses, addressing, memory hierarchy, I/O).

GATE-tagged items (2's-complement range, K-maps, flip-flop behaviour, addressable-memory & cache mapping) are Coal-India depth.

Note: original questions modelled on real exam patterns, not verbatim reproductions.